Few-Feedback Personalization of Binary Ride Preference over Road Bumps Using Hierarchical Bayesian Learning

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Abstract—**Ride preference varies across occupants, whereas most data-driven ride evaluators produce a population-level score. We present an uncertainty-aware hierarchical Bayesian framework that predicts whether an evaluator judges a road-bump response as good or bad and adapts to a previously unseen evaluator from limited feedback. Interpretable transient and frequency-weighted features are extracted from inertial and body-motion signals. Evaluator-specific logistic coefficients share a population distribution, and group spike-and-slab indicators select useful motion features. Pólya–Gamma augmentation supports posterior sampling for population learning and personalization. Projection-predictive and cross-fitted conditional-information analyses check feature sufficiency. Evaluation includes a high-feedback protocol and a sparse-feedback protocol with 10–35 labels per unseen evaluator, using discrimination, probability accuracy, log predictive density, and posterior uncertainty. In the current executable snapshot, personalization changes high-feedback-target AUROC from 0.843 to 0.846 and Brier score from 0.211 to 0.190. Across five sparse-feedback targets, macro-average Brier score improves from 0.244 to 0.205, although individual AUROC estimates remain highly uncertain. Replacing one body-motion channel with a driver-held-out recurrent reconstruction is evaluated only as a robustness study. These preliminary results motivate a confirmatory evaluator-held-out study with repeated sampling and stronger baselines.**

Index Terms—**Bayesian learning, human factors, intelligent vehicles, ride preference.**

I. Introduction

Ride quality is simultaneously a vehicle-dynamics property and a human judgment. Objective quantities such as acceleration peaks, vibration dose value, and frequency-weighted root-mean-square acceleration are useful engineering descriptors, but they do not uniquely determine whether a particular occupant likes a particular transient response. The recent review by Guastadisegni *et al.* emphasizes both the value of correlating objective measurements with subjective attributes and the need for nonlinear, multivariable models of human ride sensitivity [1]. This need becomes more consequential in intelligent vehicles, where control policies and software-defined chassis settings can in principle be adapted to the current occupant rather than tuned only for an average customer.

Three statistical difficulties arise in this setting. First, preferences differ across people, so pooling all labels into one classifier can erase meaningful heterogeneity. Second, a new occupant usually supplies only a few labels; fitting an independent model is then unstable. Third, an intelligent vehicle should distinguish uncertainty about the occupant from irreducible ambiguity in the feedback. A point estimate alone cannot indicate whether the system has observed enough feedback to personalize safely.

We address these difficulties with a hierarchical Bayesian binary-response model. Each evaluator has an individual coefficient vector, while the population mean and covariance allow partial pooling across evaluators. A previously unseen evaluator begins from the population posterior and is updated sequentially using their own good/bad feedback. Exact-zero group spike-and-slab variables, projection-predictive search, and conditional-information analysis are used to study which physically interpretable motion features support prediction. The framework targets a human-factors capability within the scope of intelligent vehicles: learning an occupant-specific ride evaluation from in-vehicle sensing and sparse interaction.

The contributions of this work are as follows.

- We formulate road-bump good/bad preference as a hierarchical Bayesian logistic model that transfers population knowledge to a new evaluator and exposes the full posterior predictive distribution during sequential personalization.
- We combine exact-zero feature inclusion with two predictive diagnostics—projection to sparse submodels and cross-fitted conditional-information gain—to separate population-level relevance, predictive redundancy, and evaluator heterogeneity among ride-response features.
- We define two evaluator-held-out protocols: a high-feedback target experiment that traces adaptation over many labels, and a sparse-feedback experiment that stresses 10–35-label target histories. Probability scores and uncertainty intervals are reported alongside ranking metrics.
- As a secondary robustness study, we replace the body bounce-rate channel with a causal recurrent reconstruction learned without the target evaluator and measure the resulting change in preference prediction.

The numerical results in this manuscript are an evidence-bound preliminary snapshot of the current implementation. They are included to make the draft executable and to expose limitations; the confirmatory study required for submission is specified in Section X.

II. Related Work

A. Objective and Subjective Ride Evaluation

Whole-body vibration standards define frequency-weighted quantities for assessing human exposure, and vehicle studies commonly use acceleration, jerk, peak, and vibration-dose descriptors [2], [3]. Such quantities are valuable priors for representation design, but an inertial measurement unit mounted near the vehicle center is not a measurement at a seat, backrest, or foot contact point. Consequently, the weighted quantities in this work are *ISO-inspired vehicle-response features*; they are not presented as standards-compliant occupant exposure values.

Machine-learning studies have related acceleration and participant characteristics to subjective ride comfort [4], and a neural evaluator has been developed specifically for speed-bump measurements [5]. These studies demonstrate that multichannel signals can predict subjective judgments, but a single pooled evaluator does not directly solve online cold-start personalization. Recent sensory-driven work explicitly accounts for differences in vibration perception through perception groups and bootstrap uncertainty [6]. Our model instead maintains a posterior coefficient distribution for each individual and adapts it from a shared continuous population distribution.

B. Preference-Aware Intelligent Vehicles

Preference-aware planning and control have been proposed to make automated-vehicle motion more consistent with occupant-specific acceleration and jerk limits [7]. Ran *et al.* use Bayesian pairwise queries to learn preferred lane-centering trajectories online [8]. These approaches establish the value of explicit preference elicitation. The present problem differs in both feedback and output: feedback is a pointwise good/bad judgment after a detected bump, and the learned quantity is a posterior evaluator model that can later serve as an input to vehicle tuning or control. Closing that control loop is outside the scope of this paper.

C. Bayesian Computation and Feature Selection

Pólya–Gamma augmentation turns the logistic likelihood into a conditionally Gaussian form, enabling efficient Gibbs updates [9]. Spike-and-slab modeling attaches posterior inclusion probabilities to candidate variables [10]. Because inclusion alone can be unstable when ride features are correlated, we also use projection-predictive selection, which seeks a small submodel that preserves a reference model's predictions [11]. Finally, conditional-information criteria help distinguish feature relevance from redundancy [12]. These tools answer different questions and are therefore reported together rather than collapsed into one importance ranking.

III. Problem Formulation

Let evaluator $u \in \{1, \ldots, U\}$ provide binary feedback for road-bump episode i. The outcome

$$y_{ui} = \begin{cases} 1, & \text{the evaluator reports a good ride response,} \\ 0, & \text{the evaluator reports a bad ride response} \end{cases} \tag{1}$$

is an operational preference label, not a standardized comfort rating. A synchronized sensor window $\boldsymbol{x}_{ui} \in \mathbb{R}^{T \times C}$ is mapped to a standardized feature vector $\boldsymbol{z}_{ui} = \phi(\boldsymbol{x}_{ui}) \in \mathbb{R}^d$, including an intercept.

The training set contains multiple evaluators with both label classes. At deployment, an unseen evaluator $\star$ initially has no personal labels. After receiving a chronological context $\mathcal{D}_{\star,t} = \{(\boldsymbol{z}_{\star i}, y_{\star i})\}_{i=1}^{t}$, the model must estimate

$$p_{\star j,t} = p(y_{\star j} = 1 \mid \boldsymbol{z}_{\star j}, \mathcal{D}_{\star,t}, \mathcal{D}_{\text{pop}}) \tag{2}$$

for held-out episodes j. The main questions are (i) how much a population prior helps at $t = 0$, (ii) how quickly personal feedback improves probability estimates, and (iii) how uncertain those estimates remain when t is small.

IV. Hierarchical Bayesian Preference Model

A. Population and Evaluator-Specific Coefficients

For evaluator u, let $\widetilde{\boldsymbol{\theta}}_u$ be a dense latent coefficient vector and let $\boldsymbol{\gamma}$ be a shared binary inclusion vector. The effective coefficients are $\boldsymbol{\beta}_u = \boldsymbol{\gamma} \odot \widetilde{\boldsymbol{\theta}}_u$. The observation model is

$$y_{ui} \mid \boldsymbol{\beta}_u \sim \text{Bernoulli}\big(\text{sigmoid}(\boldsymbol{z}_{ui}^{\mathsf{T}} \boldsymbol{\beta}_u)\big), \tag{3}$$

$$\widetilde{\boldsymbol{\theta}}_u \mid \boldsymbol{\mu}, \boldsymbol{\Sigma} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma}). \tag{4}$$

Thus, $\boldsymbol{\mu}$ captures a population tendency while $\boldsymbol{\Sigma}$ captures between-evaluator variation and correlation among preferences. The conjugate population prior is

$$\boldsymbol{\Sigma} \sim \text{InvWishart}(\nu_0, \boldsymbol{\Lambda}_0), \tag{5}$$

$$\boldsymbol{\mu} \mid \boldsymbol{\Sigma} \sim \mathcal{N}(\boldsymbol{m}_0, \boldsymbol{\Sigma}/\kappa_0). \tag{6}$$

The executable configuration sets $\boldsymbol{m}_0 = \boldsymbol{0}$, $\kappa_0 = 1$, $\nu_0 = d + 2$, and $\boldsymbol{\Lambda}_0 = \boldsymbol{I}_d$. These weak scale choices are applied after feature standardization and must be assessed by prior-predictive sensitivity checks in the confirmatory study.

B. Exact-Zero Group Spike-and-Slab Prior

Let $g = 1, \ldots, G$ index either a single feature or all features derived from one sensor. The inclusion prior is

$$\gamma_g \mid \pi \sim \text{Bernoulli}(\pi), \qquad \pi \sim \text{Beta}(a, b), \tag{7}$$

with $a = b = 1$. A unit with $\gamma_g = 0$ has an exactly zero effective coefficient for every evaluator; the intercept is always included. Posterior inclusion probability $\Pr(\gamma_g = 1 \mid \mathcal{D})$ summarizes whether a feature is supported across population draws. Because correlated features can substitute for one another, this probability is interpreted together with predictive projection and conditional-information results.

C. Pólya–Gamma Gibbs Sampling

For compactness, define $\kappa_{ui} = y_{ui} - 1/2$ and $\boldsymbol{X}_u = \boldsymbol{Z}_u \operatorname{diag}(\boldsymbol{\gamma})$. Introducing

$$\omega_{ui} \mid \widetilde{\boldsymbol{\theta}}_u \sim \text{PG}\Big(1, (\boldsymbol{\gamma} \odot \boldsymbol{z}_{ui})^{\mathsf{T}} \widetilde{\boldsymbol{\theta}}_u\Big) \tag{8}$$

makes the coefficient conditional Gaussian. With $\boldsymbol{\Omega}_u = \operatorname{diag}(\boldsymbol{\omega}_u)$, its precision and mean are

$$\boldsymbol{A}_u = \boldsymbol{X}_u^{\mathsf{T}} \boldsymbol{\Omega}_u \boldsymbol{X}_u + \boldsymbol{\Sigma}^{-1}, \tag{9}$$

$$\boldsymbol{m}_u = \boldsymbol{A}_u^{-1} \left(\boldsymbol{X}_u^{\mathsf{T}} \boldsymbol{\kappa}_u + \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu} \right), \tag{10}$$

$$\widetilde{\boldsymbol{\theta}}_u \mid - \sim \mathcal{N}(\boldsymbol{m}_u, \boldsymbol{A}_u^{-1}). \tag{11}$$

Algorithm 1 Sequential personalization of an unseen evaluator

1: Draw $\widetilde{\boldsymbol{\theta}}_\star^{(m)}$ from each population draw $(\boldsymbol{\mu}^{(m)}, \boldsymbol{\Sigma}^{(m)}, \boldsymbol{\gamma}^{(m)})$
2: Predict held-out episodes using the cold-start posterior
3: **for** $t \in \{10, 20, \dots, n_{\text{ctx}}\}$, including the final size **do**
4: Update all $\widetilde{\boldsymbol{\theta}}_\star^{(m)}$ using context $1{:}t$
5: Compute posterior predictive probabilities and uncertainty
6: **end for**
7: Report the final context result; retain the best intermediate result only as an oracle diagnostic

The sampler alternates the Pólya–Gamma variables, group indicators, inclusion rate, evaluator coefficients, and normal–inverse-Wishart population parameters. The current snapshot uses 500 burn-in iterations and retains 1500 draws without thinning.

D. Cold Start and Sequential Personalization

For posterior draw m, a new evaluator is initialized by

$$\widetilde{\boldsymbol{\theta}}_\star^{(m)} \sim \mathcal{N}(\boldsymbol{\mu}^{(m)}, \boldsymbol{\Sigma}^{(m)}), \tag{12}$$

$$\boldsymbol{\beta}_\star^{(m)} = \boldsymbol{\gamma}^{(m)} \odot \widetilde{\boldsymbol{\theta}}_\star^{(m)}. \tag{13}$$

This is the cold-start population prior. After context labels arrive, eight conditional Pólya–Gamma sweeps update $\widetilde{\boldsymbol{\theta}}_\star^{(m)}$ for every population draw while retaining the corresponding $\boldsymbol{\gamma}^{(m)}$. Predictions integrate over population and personal uncertainty:

$$\widehat{p}_{\star j,t} = \frac{1}{M} \sum_{m=1}^{M} \operatorname{sigmoid}\left(\boldsymbol{z}_{\star j}^{\mathsf{T}} \boldsymbol{\beta}_{\star,t}^{(m)}\right). \tag{14}$$

For an episode j, we summarize epistemic and Bernoulli uncertainty as

$$U_{\text{epi},j} = \operatorname{Var}_m(p_j^{(m)}), \tag{15}$$

$$U_{\text{ale},j} = \mathbb{E}_m[p_j^{(m)}(1 - p_j^{(m)})]. \tag{16}$$

The first quantity can contract as personal evidence accumulates; the second remains high for intrinsically ambiguous posterior probabilities.

V. Ride-Response Representation

A. Windowing and Preprocessing

The logger records a nominal 10-s episode at 100 Hz, spanning approximately 5 s before and 5 s after the detected bump event. The classifier uses only the interval from 2 s before to 2 s after that event. A second-order 10-Hz low-pass filter is applied before downsampling by two, yielding a 50-Hz analysis signal. Five channels are loaded: 6-D sensor pitch rate and bounce rate, and inertial vertical, longitudinal, and lateral acceleration. The current manually selected representation uses 12 statistics from bounce rate, vertical acceleration, and longitudinal acceleration; the other loaded channels remain available for ablations.

All non-intercept features are standardized with means and standard deviations computed from population-training evaluators only. For a signal $x_1, \dots, x_T$ at sampling rate f_s, the implemented transient descriptors include

TABLE I
Current Manual Feature Set

Signal	Statistics
6-D bounce rate	Absolute impulse; derivative absolute peak; derivative peak-to-peak; derivative W_k-weighted RMS
Vertical acceleration	Peak-to-peak; W_k-weighted RMS; crest factor; vibration dose value
Longitudinal acceleration	Standard deviation; W_d-weighted RMS; absolute impulse; crest factor

$$\operatorname{P2P}(x) = \max_t x_t - \min_t x_t, \tag{17}$$

$$I_{\text{abs}}(x) = f_s^{-1} \sum_t |x_t|, \tag{18}$$

$$\operatorname{Crest}(x) = \frac{\max_t |x_t|}{\sqrt{T^{-1} \sum_t x_t^2 + 10^{-12}}}, \tag{19}$$

$$\operatorname{VDV}(x) = \left(f_s^{-1} \sum_t x_t^4\right)^{1/4}. \tag{20}$$

Derivatives use a sampling-interval-aware numerical gradient. Frequency-weighted root-mean-square features multiply the discrete Fourier transform by the magnitude shape of the ISO W_k vertical or W_d horizontal weighting before inverse transformation and root-mean-square aggregation. Because the implementation does not reproduce a calibrated seat-location measurement chain, we do not interpret the resulting value as an ISO exposure measurement.

B. Complementary Feature Analyses

Projection-predictive selection treats the fitted hierarchical model as a reference. For candidate subset S and reference probability $p_{ui}^{(m)}$, it solves

$$\widehat{\boldsymbol{b}}_S^{(m,u)} = \arg\min_{\boldsymbol{b}} \frac{1}{n_u} \sum_i D_{\text{KL}}\Big[\operatorname{Bernoulli}(p_{ui}^{(m)}) \,\Big\|\, \operatorname{Bernoulli}\{\operatorname{sigmoid}(\boldsymbol{z}_{ui,S}^{\mathsf{T}} \boldsymbol{b})\}\Big]. \tag{21}$$

Forward search greedily adds a sensor or feature. The current stopping rule selects the first subset recovering 95% of the reduction from the intercept-only KL divergence to the full model, using 12 evenly spaced posterior draws.

The conditional-information diagnostic uses five within-evaluator stratified folds. At a forward step with selected set S, candidate A is scored by the cross-fitted log-score gain

$$\widehat{I}(Y; Z_A \mid Z_S, U) = \frac{1}{U} \sum_u \frac{1}{n_u} \sum_i \left[\ell_{ui}^{S \cup A, -k(i)} - \ell_{ui}^{S, -k(i)}\right]. \tag{22}$$

The auxiliary logistic models contain evaluator-specific intercepts and slopes. Equation (22) is a model-based estimate of conditional information, not a nonparametric exact mutual-information estimator.

TABLE II
EVALUATOR-HELD-OUT PROTOCOLS IN THE CURRENT EXECUTABLE SNAPSHOT

Protocol	Population evaluators	Population labels	Target evaluators	Target labels	Context/holdout construction
High-feedback target	5	2325	1	205	First 102 target labels form the growing context; the remaining 103 form a fixed holdout.
Sparse-feedback targets	5	2500	5	10–35 each (127 total)	First half of each target history is context (5–17); second half is holdout (5–18, 64 total).

VI. SECONDARY MISSING-SIGNAL ROBUSTNESS STUDY

The main preference model assumes that 6-D body bounce rate is available. To test sensitivity to that assumption, a causal sequence-to-sequence long short-term memory network [13] reconstructs bounce, roll, and pitch rates from 13 deployable channels: five inertial signals, four wheel speeds, and four motor-torque quantities. The network has one 96-unit recurrent layer and a linear three-channel output (42,915 trainable parameters). It is trained for 45 epochs using mean squared error, batch size 32, and learning rate 0.005. All normalization statistics and network weights exclude every target evaluator.

The preference pipeline is then refit after replacing measured 6-D channels in memory with the reconstruction. Although three channels are reconstructed, only reconstructed bounce rate affects the current 12-feature classifier in Table I. This experiment therefore measures robustness to a virtual bounce-rate sensor; it is not presented as a new reconstruction method or as uncertainty-aware measurement-error inference.

VII. EXPERIMENTAL PROTOCOL

A. Data and Evaluator Splits

The repository catalog contains 2891 real-vehicle episodes from 18 evaluators, of which 2682 have binary labels (1450 good and 1232 bad). The current executable experiments use 10 evaluators. Names are replaced by protocol-local identifiers, and no name is used as a model input. Table II separates the two intended use cases.

The population model never uses labels from a target evaluator. Feature scaling and feature selection are also fit only on population evaluators. Context and holdout are split in recorded order rather than randomly, making the test closer to online adaptation but leaving possible session, vehicle, location, and repeated-bump dependence to be addressed by grouped sensitivity analyses.

[TODO: Before submission, state the exact participant recruitment criteria, demographics, vehicle models/settings, road/bump protocol, feedback interface and timing, compensation, informed-consent procedure, and the formal institutional ethics determination with approval or exemption identifier. Do not claim approval until it has been issued. Finalize and justify the eligibility rule for the eight cataloged evaluators not used in these protocols.]

B. Compared Prediction Stages

For each target evaluator we report the following stages.

- *Prior*: cold-start prediction from the population posterior with zero target labels.
- *Final*: prediction after all labels in the allowed context have been assimilated.
- *Peak*: the context size with the highest holdout area under the receiver-operating- characteristic curve. This uses holdout labels for selection and is therefore an oracle diagnostic, never a deployable result.

The raw-channel and reconstructed-channel variants are trained from the same evaluator split. A complete paper must additionally compare pooled logistic regression, independent per-evaluator models where feasible, regularized partial-pooling baselines, and competitive nonlinear models under identical evaluator-held-out splits.

C. Metrics and Uncertainty

We report the area under the receiver-operating-characteristic curve (AUROC), area under the precision–recall curve (AUPRC), Brier score [14], and mean log posterior predictive density

$$\mathrm{MLPD} = \frac{1}{n}\sum_i \log\left[y_i\widehat{p}_i + (1-y_i)(1-\widehat{p}_i)\right]. \tag{23}$$

Ranking metrics assess discrimination, whereas Brier score and log predictive density penalize poor probabilities. For AUROC uncertainty, each of 600 replicates samples one posterior draw and bootstraps holdout episodes, and the 2.5th and 97.5th percentiles form an interval. This combines posterior and finite-holdout variation but does not correct dependence among episodes. Exact or stratified refitting can additionally provide pointwise expected log predictive density following Bayesian predictive-evaluation practice [15].

[TODO: Seed the new-evaluator random-number generator explicitly, run multiple independent chains and seeds, report effective sample size and rank-normalized convergence diagnostics, and replace episode bootstrap intervals with evaluator/session/bump-group-aware intervals where the data hierarchy permits.]

VIII. PRELIMINARY RESULTS

A. High-Feedback Target

Table III reports the saved run in which the unseen target provides 102 context labels and 103 episodes remain fixed for evaluation. On measured channels, personalization changes AUROC only from 0.8429 to 0.8460, but improves Brier score

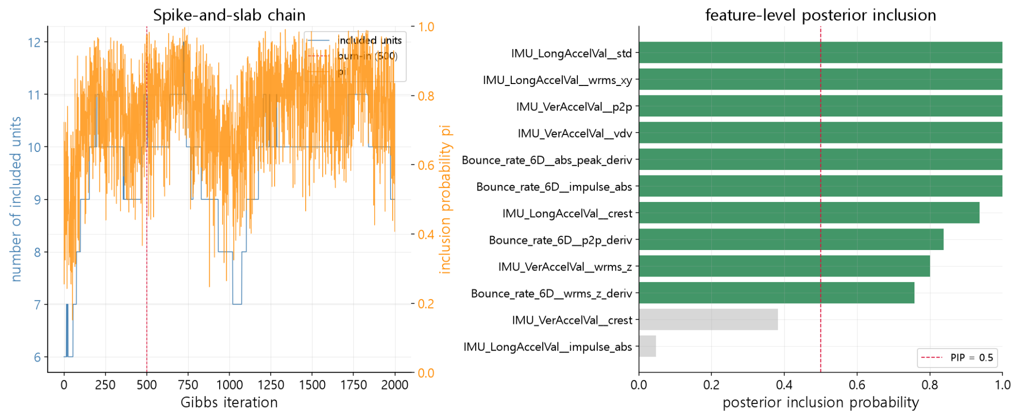

Fig. 1. Posterior inclusion probabilities from the high-feedback executable snapshot. The plot summarizes population feature support and is a diagnostic output rather than confirmatory evidence.

TABLE III
HIGH-FEEDBACK TARGET: PRELIMINARY HOLDOUT RESULTS

Input	Stage	AUROC	AUPRC	Brier	MLPD
Measured	Prior	0.8429	0.8794	0.2109	-0.6104
Measured	Final	0.8460	0.8864	0.1901	-0.5606
Reconstructed	Prior	0.8409	0.8807	0.2101	-0.6091
Reconstructed	Final	0.8560	0.8936	0.1834	-0.5461

TABLE IV
SPARSE-FEEDBACK MEASURED-CHANNEL RESULTS

Target	n_{ctx}	n_{hold}	AUROC$_0$	AUROC$_T$	Brier$_0$	Brier$_T$
S1	17	17	0.7083	0.7361	0.2115	0.2032
S2	15	15	0.6481	0.7222	0.2304	0.2474
S3	17	18	0.7000	0.4625	0.2404	0.2520
S4	5	5	1.0000	0.8333	0.1880	0.1842
S5	9	9	0.3750	0.7500	0.3500	0.1356
Macro	–	–	0.6863	0.7008	0.2441	0.2045

from 0.2109 to 0.1901 and mean log predictive density from -0.6104 to -0.5606. Thus, in this split, the clearest benefit is better probability quality rather than a large change in ranking. The joint posterior–bootstrap AUROC interval is $[0.747, 0.917]$ for the final measured-channel model; the cold-start interval is much wider, $[0.207, 0.889]$, because population-posterior draws disagree about the target ranking.

The recurrent virtual sensor attains median held-out waveform correlations of 0.992, 0.995, and 0.989 for bounce, roll, and pitch rate, respectively. Its bounce-feature correlations range from 0.859 for derivative absolute peak to 0.992 for absolute impulse. The reconstructed preference variant is not worse in this single split and slightly improves all four final metrics, but the difference is too small and too dependent on one target to support a superiority claim.

[TODO: Regenerate the target-adaptation uncertainty plot with anonymous evaluator identifiers, publication-size typography, and fixed feedback checkpoints before including it in the final paper.]

B. Sparse-Feedback Targets

Table IV shows the five sparse targets using measured channels. The final update uses only 5–17 context labels, and holdouts contain as few as five episodes. Macro-average AUROC rises from 0.6863 to 0.7008 and Brier score improves from 0.2441 to 0.2045. However, target S3 becomes worse in AUROC after adaptation, and the near-perfect estimates for S4 or S5 are based on very few positive–negative pairs. The result therefore supports neither universal improvement nor a precise estimate of the mean effect. It instead demonstrates the regime in which posterior uncertainty and proper probability scores are essential.

For the reconstructed variant, macro Brier score similarly improves from 0.2438 to 0.2069 and mean log predictive density from -0.6823 to -0.5988, while macro AUROC falls from 0.6986 to 0.6368. This mixed result is consistent with the intended interpretation of reconstruction as a robustness stress test rather than a performance contribution.

C. Population Feature Structure

Table V presents representative measured-channel posterior summaries from the high-feedback split. Vertical acceleration peak-to-peak and longitudinal weighted RMS are the first two features selected by the two predictive diagnostics, although their coefficient signs are conditional on the correlated standardized design. The projection search requires seven features plus the intercept to recover 95.4% of the full-model KL reduction. The forward conditional-information analysis retains six features before the next gain becomes nonpositive.

The posterior mean signs must not be read causally. Road severity, speed, vehicle setting, bump identity, and evaluator history may be confounded, and correlated transient statistics can reverse conditional signs. The defensible result is that several features jointly preserve predictive structure while evaluator-specific coefficients vary; causal suspension-design statements require controlled interventions.

TABLE V
REPRESENTATIVE POPULATION FEATURE RESULTS IN THE HIGH-FEEDBACK SPLIT

Feature	PIP	Mean coefficient	95% interval	Forward CMI (nat)	Diagnostic role
Vertical acceleration peak-to-peak	1.000	-0.694	$[-1.317, -0.104]$	0.07148	First in projection and CMI
Longitudinal acceleration W_d RMS	1.000	-0.647	$[-1.407, 0.094]$	0.01449	Second in CMI; fifth in projection
Longitudinal acceleration standard deviation	1.000	0.564	$[-0.094, 1.189]$	0.00487	Third in CMI; sixth in projection
Bounce-rate absolute impulse	1.000	0.600	$[0.105, 1.094]$	0.00318	Early projection representative
Bounce-rate derivative W_k RMS	0.758	-0.155	$[-0.831, 0.372]$	0.00525	Later nonredundant CMI gain
Vertical acceleration vibration dose	1.000	-0.070	$[-0.514, 0.387]$	–	Completes 95% projected subset

IX. DISCUSSION

A. What Personalization Changes

The high-feedback split suggests that the population prior already ranks episodes reasonably well, whereas target feedback sharpens probability calibration and predictive density. This is a useful distinction for intelligent-vehicle applications. A controller or recommendation layer often needs a probability that reflects confidence, not only a ranking. The hierarchical prior avoids fitting a new model from scratch, and the posterior trajectory makes it possible to define a future stopping rule based on uncertainty rather than selecting the context with best holdout AUROC.

The sparse-target results show the other side of partial pooling. A few observations can move the posterior in a helpful direction, but they can also be unrepresentative. Strong shrinkage toward the population is desirable until the target supplies both good and bad examples. Future deployment should therefore expose an abstention or "request more feedback" state and should not treat the oracle peak as an online early-stopping mechanism.

B. Interpretability and Sensor Robustness

The three feature analyses are complementary. Spike-and-slab inclusion asks whether a feature is active across posterior draws. Projection asks how compactly the full fitted predictions can be reproduced. Cross-fitted conditional information asks whether a candidate improves held-out log score after conditioning on the already selected features and evaluator identity. Agreement on vertical peak-to-peak and longitudinal weighted RMS makes them stable descriptive candidates, but not necessarily physical causes of good or bad judgments.

The reconstruction study indicates that a high-fidelity virtual bounce-rate signal can preserve downstream probability quality in one held-out target. A rigorous missing-sensor treatment would propagate reconstruction uncertainty into the preference likelihood rather than substituting a point waveform. That extension is intentionally separated from the paper's present contribution.

C. Limitations

The number of evaluators is small relative to the number of episodes, and episode-level sample size does not replace evaluator-level replication. Several sparse holdouts contain too few examples of one class for stable AUROC inference. The current split does not explicitly group repeated passes by bump, location, session, vehicle, or suspension setting, so within-group dependence and distribution shift may be understated. Good/bad feedback is intentionally simple but cannot be equated with a validated multi-attribute ride-comfort scale. The feature weights are vehicle-response proxies rather than calibrated whole-body exposure measurements. Finally, the current artifact uses one population chain and one principal split, and the available comparison set is not yet sufficient for a journal claim of superiority.

X. CONFIRMATORY STUDY REQUIRED FOR SUBMISSION

The final T-IV evaluation should freeze preprocessing, priors, inclusion level, and stopping rules before inspecting test results. At minimum, it should include the following.

1) Leave one sufficiently sampled evaluator out in turn, fitting every normalization, feature-selection step, reconstruction model, and population posterior without that evaluator.
2) For sparse feedback, subsample context budgets such as 0, 2, 5, 10, and 20 with repeated chronological starting points or session-aware blocks; keep a fixed group-held-out test set.
3) Compare against a population-only logistic model, regularized independent models, mixed-effects or empirical-Bayes logistic regression, and at least one nonlinear sequence or tree-based baseline under the identical outer split.
4) Report evaluator-macro proper scores with evaluator-level confidence intervals, calibration curves, sensitivity to class imbalance, prior-predictive checks, and multiple-chain convergence diagnostics. Treat episode-level intervals as secondary.
5) Repeat the measured-versus-reconstructed comparison across outer folds and propagate or bound reconstruction error; do not tune the reconstruction network on final target evaluators.

[TODO: Replace all preliminary tables and the abstract's numerical claims with results from the frozen confirmatory pipeline. Record software versions, hardware, run identifiers, wall-clock costs, and inference seeds in a reproducibility appendix.]

XI. CONCLUSION

This paper develops a hierarchical Bayesian framework for predicting evaluator-specific good/bad ride preference after road bumps. The population posterior supplies a principled cold start, and Pólya–Gamma personalization updates both predictions and uncertainty as sparse target feedback arrives. Exact-zero inclusion, predictive projection, and conditional-information analysis retain the interpretability of motion-derived features. Current executable results show improved probability scores for one high-feedback target and on average across five sparse targets, while also revealing substantial evaluator-level uncertainty and occasional negative adaptation. A driver-held-out recurrent reconstruction preserves performance in the high-feedback snapshot but remains a secondary robustness result. The next step is not a broader claim from the present split; it is the confirmatory evaluator- and session-grouped evaluation specified above.

Acknowledgment

[TODO: Add funding, data-collection acknowledgments, conflicts of interest, and any required ethics/consent statement after institutional confirmation.]

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[TODO: Add IEEE-format author biographies and photographs for the final files.]